

```
def strassen(A, B):
```

```
    a = A[0][0]
    b = A[0][1]
    c = A[1][0]
    d = A[1][1]
```

```
    e = B[0][0]
    f = B[0][1]
    g = B[1][0]
    h = B[1][1]
```

```
    p1 = a * (f - h)
    p2 = (a + b) * h
    p3 = (c + d) * e
    p4 = d * (g - e)
    p5 = (a + d) * (e + h)
    p6 = (b - d) * (g + h)
    p7 = (a - c) * (e + f)
```

```
    C11 = p5 + p4 - p2 + p6
    C12 = p1 + p2
    C21 = p3 + p4
    C22 = p1 + p5 - p3 - p7
```

```
    return [[C11, C12], [C21, C22]]
```

```
A = [[1, 2],
      [3, 4]]
```

```
B = [[5, 6],
      [7, 8]]
```

```
C = strassen(A, B)
```

```
print("Matrix A:")
print(A)
```

```
print("Matrix B:")
print(B)
```

```
print("Result:")
for row in C:
    print(row)
```

Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:00) on win32

Enter "help" below or click "Help" above for more info.

```
>>>
```

```
==== RESTART: C:/Users/ROOPA SAI DIVYA/OneDrive/Docume
```

```
Matrix A:
```

```
[[1, 2], [3, 4]]
```

```
Matrix B:
```

```
[[5, 6], [7, 8]]
```

```
Result:
```

```
[19, 22]
```

```
[43, 50]
```

```
>>>
```